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Effort Based Performance Measurement in Business Process Management

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This paper presents a holistic methodology and associated toolset for business process modelling and analysis. The importance of integrating performance measurement (PM) indicators and tools in business process management has been identified in the literature as well as contemporary approaches that attempt to give a solution to this problem. However, most of these approaches propose solutions based on theories and techniques used in other disciplines or research fields which approach the issue of performance measurement in BPR from their individualistic perspective. In the majority of cases the proposed theories and techniques prove to be inadequate to measure performance holistically and integrate, organizational, human resource management, process management and workflow management concepts. The resulting methodology and toolset is based on business analysis ratios that *define* and *measure* key performance indicators in qualitative and quantitative terms and identify the relationships between them. These ratios allow enterprises to achieve a performance evaluation tool for continuous improvement in their attempt to respond and adapt quickly to turbulent markets. The performance measurement toolset (called ADJUST) is used for (1) planning different reorganization scenarios for the achievement of desired performance, (2) for assessing real time performance and (3) reporting deviations from desired planned performance.

In order to monitor and assess real time performance the ADJUST software tool provides interfaces to three kinds of management tool categories, namely:

- *Process management tools (PMT)*: business models, cycle time, primitive cost, etc.
- *Human resource management (HRM) tools*: job descriptions, performance measures, etc.
- *Workflow management (WFM) tools*: transactions, business rules, workflow models, etc.

The ADJUST business analysis engine is based on employee effort analysis and provides meta-analysis of data analysed in PMT and HRM tools. Data like activity sequencing, cycle times, primitive costing, etc. from PMT tools are analysed in ADJUST in order to assess the percentage of high-business and low value added activities, mission-non mission activities, concentration analysis, etc. Knowledge management elements are integrated to job descriptions and associated with performance measures and workflow elements. Similarly, job descriptions and performance measurement data that are provided by HRM tools are analysed further in terms of cost to manage, principal-agent analysis etc. In the case of WFM tools the ADJUST system provides not only meta-analysis of workflow processes, transactions and document flows but also dynamic monitoring of their performance in real time. Copyright © 2010 John Wiley & Sons, Ltd.

INTRODUCTION AND STATE OF THE ART IN BUSINESS PROCESS MODELLING

Because business processes are complex, process designers will often provide different modelling *views*, each focussing on one aspect of the process. Curtis identified four views, summarized below (Curtis *et al.*, 1992):

- *The functional view*: this view presents the functional dependencies between the process elements (activities, subprocesses, etc.). These dependencies are typically embodied in the fact that some process elements consume (need) data (resources) produced by others. Typical notations used in the functional view include event Process Chain Diagrams (ePCD), IDEF1-2, etc.
- *The dynamic (behavioural) view*: the dynamic view provides sequencing and control information about the process, when certain activities are performed (timing, pre-conditions) and how they

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are performed, i.e. state transition diagrams, petri nets, etc.

- *The informational view*: this view includes the description of the entities that are produced, consumed or otherwise manipulated by the process. These entities include pure data, artefact, products, etc.
- *The organizational-structural view*: this view describes *who* performs a given task or function, and *where* in the organization (functionally and physically).

As is common with modelling methods, different notations may be appropriate for different views.

Many other approaches have introduced other perspectives to BPR (Zelm *et al.*, 1995; Teng *et al.*, 1996). For a performance framework to be considered as holistic has to be providing tools, techniques and ratios that fall to most of these categories (Adam *et al.*, 1997).

The 10 perspectives in business analysis techniques and ratios

Tools and techniques identified in the literature are usually classified these four perspectives. We have expanded the number of these four perspectives to include issues like strategic alignment, change management, customer requirements, performance measurement, organizational drivers, etc. in our effort to classify tools and techniques used in business analysis in the literature (Neely *et al.*, 1996; Jagdev *et al.*, 1997; Rolstadas, 1998; Valiris and Glykas, 1999; Tonchia, 2000; Van Der Merwe, 2002; Xirogiannis *et al.*, 2004; Reijers and van der Aalst, 2005; Trienekens *et al.*, 2005; Gruat La Forme *et al.*, 2007; Nudurupati *et al.*, 2007; Škerlavaj *et al.*, 2007; Tan *et al.*, 2007; Turetken and Schuff, 2007; Elbashir *et al.*, 2008; Footen and Faust, 2008; Gregoriades and Sutcliffe, 2008a; Jazayeri and Scapens, 2008; Vergidis *et al.*, 2008; Cheng *et al.*, 2009; Han and Park, 2009; Han *et al.*, 2009; Herzog *et al.*, 2009; Jeston and Nelis, 2009; Lam *et al.*, 2009; Li *et al.*, 2009; Yen, 2009; Trkman *et al.*, 2010; Trkman, 2010). Each of these 10 perspectives is described briefly below:

- (1) *Opportunity assessment*: the processes for methodically identifying, focusing, prioritizing and resolving issues and opportunities related to implementing change within an organization. Analytical tools within this cluster help the consultant to brainstorm ideas, prioritize issues and opportunities as well as assess the impact of change.
- (2) *Activities*: how effort is spent within an organization at an individual, functional, process, product, or service line level. Analytical tools within this tool cluster are designed to estimate the value, efficiency and/or effectiveness of individual and organizational effort.

- (3) *Processes*: the horizontal linkage of activities needed to achieve a desired result. Analytical tools within this tool cluster are designed to reengineer, redesign, streamline and optimize business processes.
- (4) *Structure*: the vertical linkage of organizational functions, processes and resources within an organization. Analytical tools within this tool cluster assess organizational structure issues including the efficiency of reporting relationships, the alignment of functions and the orientation of resources.
- (5) *Requirements*: the customer's needs, requirements and priorities that must be fulfilled to achieve optimum organizational performance. Analytical tools within this tool cluster identify where resources need to be realigned to meet existing and future requirements.
- (6) *Outputs*: the internal and external products or services produced by an organization. Analytical tools within this tool cluster assess and challenge the internal and external demands for products and services.
- (7) *Drivers*: the policy, structural and operational factors, which impact and influence the operations and cost of an organization. Analytical tools within this tool cluster help the consultant identify, determine the impact of, and validate the appropriateness of various drivers.
- (8) *Inputs*: the human resources and other assets required to produce products or services. Analytical tools within this tool cluster help determine the level, mix and deployment of resources within the organization.
- (9) *Performance measurement*: The metrics of success or failure, which influence the operations of an organization. Performance measure analyses can be related to financial performance, quality and service levels, operational performance and/or productivity.
- (10) *Change management*: the processes for managing an organization through implementation. The analytical tools within this tool cluster identify and manage factors, which may hinder as well as help to ensure implementation success.

These tool clusters are used in conjunction with one another to achieve a complete view of the organization. A list of tools used per cluster are described in the table below:

Review of techniques and ratios used in different business situations

The aforementioned techniques and tools can be classified and according to the business situations BPR projects attempt to analyse. Typical business situations resolved by BPR projects are presented below.

Activities	Processes	Structure
Mission/Non-Mission (Primary/Secondary) Activity Value Added Analysis Cost of Quality Analysis Fragmentation/Concentration Analysis Fractionalization Analysis Equivalent Salary Analysis Work Expectancy Development	Process Prioritization Structured Process Selection Reengineering Master Planning Visioning Core process Analysis & Design Cross Functional Process Analysis & Design Detailed Process Flow Analysis & Design Process Profiling Brown Paper Analysis Process Critical Path Analysis Process Control Risk Assessment Methods Analysis Cycle Time Reduction Analysis Approval Cycle Analysis Policy and Procedure Definition	Management Structure Analysis Internal Contact Survey Analysis Activity Clustering Detailed Organizational Design Accountability Framework Job Description Definition
Opportunities Assessment	Inputs	Change Management
Customer Attribute Baseline Ranking Customer Attribute Categorization Current vs. Future Requirements Analysis Client vs. Competitor Analysis External vs. Internal Perceptions Analysis Acceptable vs. Outstanding Analysis Internal Requirements Assessment Implementation Action Planning Management Contracts Implementation Scorekeeping	Staffing Trend Analysis Staffing Turnover Analysis Resource to Workload Matching Skill Mix Analysis Make vs. Buy Analysis	Communication Planning Culture Print Change Propensity Analysis Force Field Analysis Attrition Analysis Position Elimination Scoring Relocation/Retraining Analysis
Performance Measures	Outputs	Problem Solving
Performance Measures Framework Financial Analysis Benchmarking Best Practices Analysis Economic Decomposition Modelling	Output Rationalization Activity Based Product/Service Line Costing Customer Profitability Analysis Channel Strategy Analysis	Problem Solving Process Interviewing 'What One Thing' Analysis Opportunity Logging Cause and Effect Analysis Brainstorming
Driver		Delphi Technique Nominal Group Technique List Reduction Selection Grid Weighted Voting Paired Comparisons Impact Analysis

- *Restructuring*. Facilitates refocusing of the organization to meet key strategic or customer-driven goals. Specific restructuring situations include:
 - Mismatch between organizational structure and business strategies.
 - Inadequate customer or market focus.
 - Mismatch of resources to functional workload requirements.
 - Decreasing productivity and efficiency.
- *Rightsizing*. Typically focuses on cost management. Specific rightsizing situations include:
 - Declining business volume.
 - Inadequate profitability.
 - Need to reduce pricing structure.
- *Uncompetitive cost structure*.
- Excessive layers of management.
- Increasing demand for services.
- Decreasing productivity and efficiency.
- *Reengineering*. Focuses on redesigning and streamlining cross-functional business processes. Specific reengineering situations include:
 - Cumbersome business processes and systems.
 - Inadequate quality focus.
 - Backlogged processes or workload.
 - Inadequate service levels.
 - Mismatch of resources to process workload requirements.
 - Decreasing productivity and efficiency.

- **Integration.** Focuses on achieving synergies among merged or acquired entities. Specific integration situations include:
 - Merger or acquisition of another company.
 - Need to consolidate divisions or business units.

The table below lists the methods and tools, which are typically used in each of these business situations. As shown, many tools are applicable to more than one business situation. The three symbols used present high (●) medium (⊙) and low (○) applicability:

METHOD-TOOL CLUSTER	TOOL	BUSINESS SITUATION			
		Rightizing	Reengineering	Restructuring	Integration
PROBLEM SOLVING					
	Problem Solving Process	●	●	●	●
	Interviewing	●	●	●	●
	"What One Thing" Analysis	●	●	●	●
	Opportunity Logging	●	●	●	●
	Cause and Effect Analysis	⊙	●	⊙	⊙
	Brainstorming	●	●	●	●
	Delphi Technique	⊙	⊙	⊙	⊙
	Nominal Group Technique	⊙	●	●	⊙
	List Reduction	●	●	●	●
	Selection Grid	⊙	⊙	⊙	⊙
	Weighted Voting	⊙	⊙	⊙	⊙
	Paired Comparisons	⊙	⊙	⊙	⊙

	Impact Analysis	●	●	●	●
ACTIVITIES	Mission/Non-Mission (Primary/Secondary)	●	●	●	●
	Activity Value-Added Analysis	●	●	●	●
	Cost of Quality Analysis	⊙	●	⊙	⊙
	Fragmentation/ Concentration Analysis	●	●	●	●
	Fractionalization Analysis	●	⊙	●	⊙
	Equivalent Salary Analysis	●	⊙	●	⊙
	Work Expectancy Development	●	⊙	⊙	⊙
PROCESSES		Rightizing	Reengineering	Restructuring	Integration
	Process Prioritization	⊙	●	⊙	⊙
	Structured Process Selection	○	●	○	○
	Reengineering Master Planning	○	●	○	○
	Visioning	⊙	●	⊙	⊙

	Core Process Analysis & Design	⊙	●	⊙	●
	Cross-Functional Process Analysis & Design	⊙	●	⊙	●
	Detailed Process Flow Analysis & Design	⊙	●	⊙	⊙
	Process Profiling	⊙	●	⊙	⊙
	Brown Paper Analysis	○	●	○	○
	Process Critical Path Analysis	○	●	○	○
	Process Control Risk Assessment	○	●	○	○
	Methods Analysis	⊙	●	⊙	⊙
	Cycle Time Reduction Analysis	○	●	⊙	⊙
	Approval Cycle Analysis	⊙	●	⊙	⊙
	Policy and Procedure Definition	⊙	●	●	⊙
STRUCTURE	Management Structure Analysis	●	⊙	●	●
	Internal Contact Survey Analysis	○	⊙	●	⊙
	Top-Level Organizational Design	⊙	⊙	●	●
	Activity Clustering	⊙	⊙	●	⊙
	Detailed Organizational Design	⊙	⊙	●	⊙
	Accountability Framework	●	●	●	●
	Job Description Definition	⊙	●	●	●
REQUIREMENTS	Customer Attribute Baseline Ranking	⊙	●	⊙	⊙
	Customer Attribute Categorization	⊙	●	⊙	⊙
	Current vs. Future Requirements Analysis	⊙	●	⊙	⊙
	Client vs. Competitor Analysis	⊙	●	⊙	⊙
	External vs. Internal Perceptions Analysis	⊙	●	⊙	⊙
	Acceptable vs. Outstanding Analysis	⊙	●	⊙	⊙
	Internal Requirements Assessment	⊙	●	⊙	⊙
OUTPUTS		Rightizing	Reengineering	Restructuring	Integration
	Output Rationalization	●	●	○	○

	Activity-Based Product/Service Line Costing	●	●	⊙	⊙
	Customer Profitability Analysis	●	●	⊙	⊙
	Channel Strategy Analysis	●	●	⊙	⊙
DRIVERS	Macro Cost Driver Analysis	●	●	●	●
	Activity Generator Analysis	●	●	●	●
	Time Driver Analysis	○	●	⊙	⊙
INPUTS	Staffing Trend Analysis	●	○	⊙	⊙
	Staffing Turnover Analysis	●	⊙	⊙	⊙
	Resource to Workload Matching	●	⊙	●	⊙
	Skill Mix Analysis	●	●	●	●
	Make vs. Buy Analysis	⊙	⊙	●	⊙
PERFORMANCE MEASURES	Performance Measures Framework	●	●	●	●
	Financial Analysis	●	⊙	●	⊙
	Benchmarking	●	●	●	●
	Best Practices Analysis	●	●	●	●
	Economic Decomposition Model	●	⊙	●	⊙
CHANGE MANAGEMENT	Communication Planning	●	●	●	●
	Culture Print	○	⊙	●	●
	Change Propensity Analysis	⊙	⊙	⊙	⊙
	Force Field Analysis	⊙	●	⊙	⊙
	Attrition Analysis	●	○	⊙	○
	Position Elimination Scoring	●	○	○	○
	Relocation/ Retaining Analysis	●	⊙	●	⊙
	Implementation Action Planning	●	●	●	●
	Management Contracts	●	⊙	⊙	⊙
	Implementation Scorekeeping	●	⊙	⊙	⊙

PERFORMANCE MEASUREMENT TECHNIQUES AND TOOLS IN BPR

There are many contemporary approaches that introduce performance measurement techniques

and tools in BPR but the vast majority of them use a specific technique or theory usually developed in other research fields.

The most widely used techniques and theories used for BPR performance measurement are based on the Analytical Hierarchy Process (Yen, 2009), contingency theory (Trkman, 2010), micro and macro process analysis (Han *et al.*, 2009), quantitative methods based on time series intervention (Lam *et al.*, 2009), Quality Function Deployment (Jagdev *et al.*, 1997), activity based costing and activity based management (Tan *et al.*, 2007; Grahovac and Devedzic, 2010), queuing theory (Cheng *et al.*, 2009), structural equation modelling (Reijers and van der Aalst, 2005; Škerlavaj *et al.*, 2007; Trkman *et al.*, 2010) strategic analysis based on goal, business objective and critical success factors linking with processes with the use of strategy maps (Trienekens *et al.*, 2005; Nudurupati *et al.*, 2007; Trkman, 2010), collaborative supply chain techniques (Reijers and van der Aalst, 2005; Gruat La Forme *et al.*, 2007), employee performance measurement (Gregoriades and Sutcliffe, 2008a), human performance shaping factors (Gregoriades and Sutcliffe, 2008a, b), Bayesian Belief Networks (Gregoriades and Sutcliffe, 2008a), process cycle time analysis (Gregoriades and Sutcliffe, 2008a), systems theory (Van Der Merwe, 2002; Ng, 2004), risk theory in Business Process Outsourcing (Gewald and Dibbern, 2009) and audit risk assessment (Carnaghan, 2006; Dunn, 2006; Durigon, 2006), business rules (Muehlen and Indulska, 2010; Lee *et al.*, 2007; Gašević *et al.*, 2010).

Many researchers have indicated the need for holistic performance measurement approaches in BPR and the need for linkages between the two subjects (Neely *et al.*, 1996; Škerlavaj *et al.*, 2007; Elbashir *et al.*, 2008; Herzog *et al.*, 2009; Jeston and Nelis, 2009).

The need also stems from organizational performance measurement researchers who argue that existing methodologies and tools fail to integrate concepts and tools from process management, human resource management and workflow and lack business analysis metrics and ratios, which would link strategic performance indicators with organizational and employee objectives (individualistic performance), and manipulate qualitative measurements (Hansen, 1994; Manganelli and Klein, 1994; Becker and Gerhart, 1996; Bititci *et al.*, 1997).

Additionally, existing performance measurement methodologies and tools present the following limitations:

- Are difficult to be used as an integral part of decision making at either strategic or operational levels (Ahmed *et al.*, 1996; Bourne *et al.*, 2000);
- Do not encourage employee participation and creative thinking. Moreover they do not show

how planned improved performance can be realised (Lalli, 1996; Deming, 1986; Bititci *et al.*, 2000);

- Do not link properly with simulation, analytical and workflow tools. This makes business performance difficult to be quantified and to be integrated in information systems (Ahmed *et al.*, 1996; Lalli, 1996);
- Depend for successful application to a large extent on the support of external consultants, which jeopardises continued success after the consultants leave the company.

Classification of tools related to performance management

Data warehouses are databases for historical transaction data. Users can typically store and analyse data on product mixes and sales that are keyed according to various criteria. Generic data warehouses must be adapted by the customer to the source system; no business management informational content is supplied (Bradley *et al.*, 1995).

With ERP-specific solutions, predefined structures are supplied along with the data warehouse, but just as with the above-mentioned solution, they do not provide an interface for process data.

Executive Information Systems are modular systems that are used to navigate through datasets stored in data warehouses or databases. The user must create his or her own analyses and reports. Specialized visualization systems such as Management Cockpit and Balanced Scorecard systems take this approach one step further. These offer partially predefined navigational structures for datasets. One thing that all of the aforementioned approaches have in common is that they do not supply any information on process efficiency, such as cycle times and organizational gaps.

Sales information systems and other analysis systems specially conceived for specific functional areas are a combination of the previously mentioned data management and visualization systems. They use transaction data from application systems and offer standardized display formats for these data. Because they lack suitable collection procedures, however, they do not provide any performance data.

A special group of application systems works with process-related performance data. These are the activity-based cost calculation systems. Activity based costing (ABC) systems permit the calculation of cost factors based upon measured or estimated throughput times. This provides interesting information for planning and controlling purposes. These systems present weaknesses as they do not support data collection, and thus can result in a great deal of expense and the possibility of measurement errors. In cases deviations from

desired performance are detected, no channels are provided for feedback to the participants for the purposes of initiating measures for improvement.

While many companies have now implemented these expensive and far-reaching software systems via ERP solutions, they typically lack reliable information feedback of business performance in real time and deviation from planned performance. How, then, can companies ensure that they have an appropriate level of control over the business processes that are being enabled by those applications? And how can they successfully address the need for longer range planning, in areas such as implementing an increasing span of e-business capabilities? (O'Mara *et al.*, 1998; Valiris and Glykas, 1999).

More specifically, companies need the ability:

- To verify that scheduled process throughput times are being met (and how consistently those times are met).
- To detect weak points in their processes, where bottlenecks or malfunctions are likely to occur.
- To detect points where a potential for time and/or cost-saving improvements exist.
- To verify that process capacities correspond accurately to the amount of work.
- That needs to be accomplished to meet customer demands.
- To verify that proposed process costs are being adhered to.
- To ascertain whether planning premises were correct.
- To ensure that the business processes currently being implemented are resulting in desired levels of customer satisfaction.

THE ADJUST TOOL

Existing business process methodologies and modelling languages come from different scientific traditions, and as such, serve different purposes and represent different things (Curtis *et al.*, 1992). Ould argues that business process modelling is useful for three basic reasons (Ould, 1995):

- (1) *Describing a process*: we model a process to be able to describe it. We could have two different audiences for these descriptions: (a) humans, in which case understandability is important (Curtis *et al.*, 1992), or (b) machines, in which case formality is important.
- (2) *Analysing a process*: process analysis consists of assessing the *properties* of a process. Process re-engineering and improvement relies on an analysis of existing processes to identify redundant or sub-optimal steps. If the process is described *formally*, we can *verify mechanically* structural properties such as coupling and

cohesion (Phalp and Shepperd, 2000) or dynamic properties such as the absence of deadlock, liveness properties, and the like. We can also measure a process and identify gaps between current and desired performance.

- (3) *Enacting a process*: we may enact a process for simulation purposes, or to provide some level of support for process execution. Depending on the language, this support can range from reacting to the execution of the activities of the process to check that specific constraints are satisfied, to actually driving the execution of the process (Curtis *et al.*, 1992). Only formal languages make process enactment possible (Dreiling *et al.*, 2008).

Business Process Modelling Languages (BPML) describe different aspects of processes (organization, activities and processes, job descriptions, data items, etc). Process modelling is the first step in every BPR project (Reijers and Mansar, 2005; Decker and Mendling, 2009; Grahovac and Devedzic, 2010; Muehlen and Indulska, 2010; Rodríguez *et al.*, 2010). Business process models are abstract models which contain key information about the tasks that are being carried out to achieve the company's goals. Modelling notations that are used by BPR tools and are close to executable business process models are based on the Unified Modelling Language (UML) (Lima *et al.*, 2009; Woodside *et al.*, 2009; Lassen and van der Aalst, 2009; Noguera *et al.*, 2010; Garousi, 2010), and the Business Process Modelling Notation (BPMN) (Harmon and Davenport, 2007; Dijkman *et al.*, 2008; Lassen and van der Aalst, 2009; Wong and Gibbons, 2009a; Wong and Gibbons, 2009b). Event Process Chain (EPC) diagrams (Kindler, 2006; Mendling *et al.*, 2008; Bevilacqua *et al.*, 2009; Decker and Mendling, 2009; Lassen and van der Aalst, 2009).

Recently many researchers have integrated BPR with workflow concepts (Liu and Shen, 2003; Reijers and Mansar, 2005; Huang *et al.*, 2006; Lee *et al.*, 2007; Shen *et al.*, 2007; Dreiling *et al.*, 2008; Decker and Mendling, 2009; Lassen and van der Aalst, 2009; Jiang *et al.*, 2010; Weber *et al.*, 2010). Decker and Mendling (2009) elaborate on the importance of process instantiation after identifying the lack of precise semantics both on pragmatic terms and theoretical underpinnings. Shen *et al.* (2007) argue that the integration of Web services into BPR is a recent outgrowth of the Business Process integration field that will require powerful meta-schema matching mechanisms supported by higher level abstractions, such as UML meta-models. Currently, there are many XML-based workflow process specification languages (e.g. XPD, BPML) which can be used to define business processes in the Web services and Grid Computing world. However, with limited capability to describe the relationships (schemas or ontologies) between process objects, the

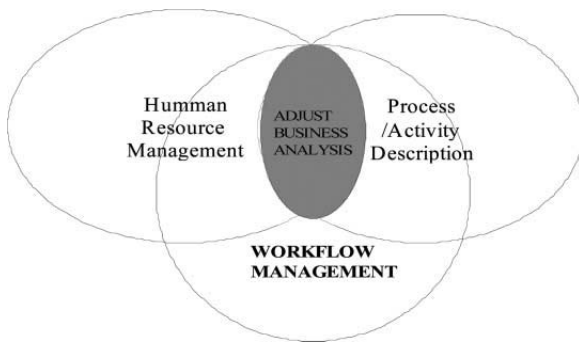


Figure 1 The Adjust Approach.

dominant use of XML as a meta-data markup language makes the semantics of the processes ambiguous.

In ADJUST we also recognise the issue and we argue that the way forward is by integrating the web or stand alone BPR tools with the online ADJUST business analysis engine.

The main focus of process related performance measures in ADJUST is the analysis of the business models with a view of developing solutions for redesign. The emphasis is on *why* the operations and processes both at the individualistic and holistic view of the organization are performed the way they are performed (Tonchia, 2000; Valiris and Glykas, 2000; Glykas and Valiris, 1999; Valiris and Glykas, 1998).

Business process analysis has rapidly gained appreciation as the appropriate approach to research and practice in the management of organizations in the increasingly competitive environment of the 1990s and beyond. The emergence of business process analysis as a new paradigm for management research is a result of a coming together of a wide range of different strands of thinking, rather than a coherent theoretical development of business process concepts (Ould, 1995; Glykas and Valiris, 1998; Glykas and Valiris, 1999).

The Adjust approach is presented in the Figure 1.

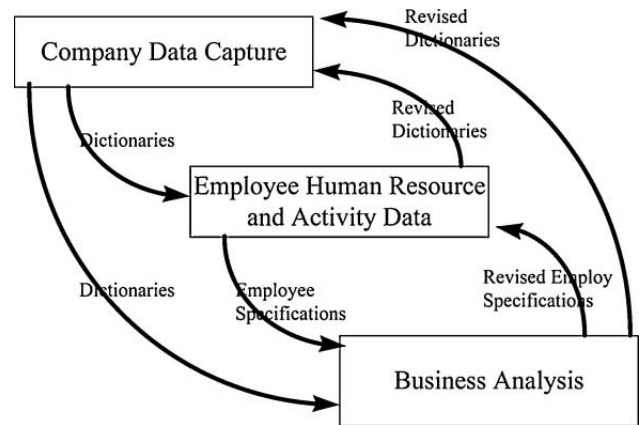


Figure 2 Integration of Individualistic and Holistic Views.

The aim of the approach is to integrate human resource management and workflow management with business process reengineering concepts in order to perform business analysis that applies both to the holistic (process) and individualistic (employee) view of the organization. This is described in the Figure 2.

The holistic perspective of the organization is described in the company data capture module, where company wide management data in the form of dictionaries are held. The individualistic perspective of the organization is described in the employee description module, where employee specific data based on the dictionaries provided by the previous module are specified. The final business analysis module provides analytical tools and techniques for achieving the desired performance both in terms of employee and dictionary specifications.

A schematic representation of ADJUST is presented in the Figure 3.

The ADJUST software platform is shown in Figure 4. It provides the user with a number of tools that store and analyse information stored into repositories belonging to different satellite tools. This information is transferred into the ADJUST

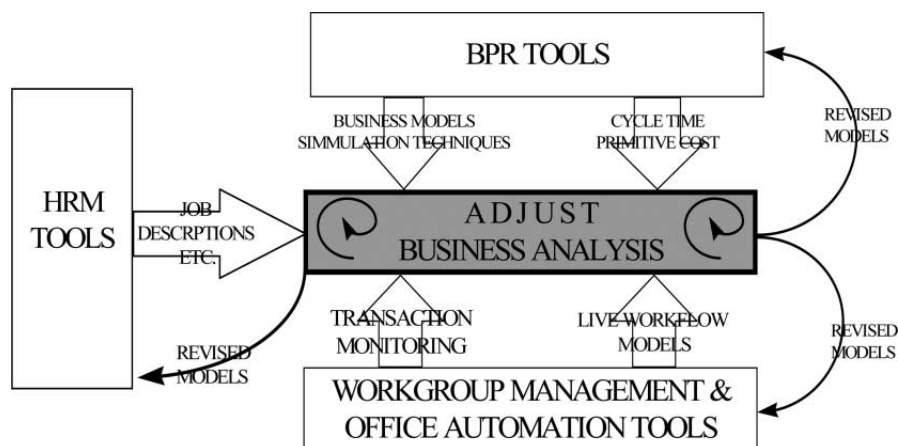


Figure 3 The Adjust Business Analysis Engine.

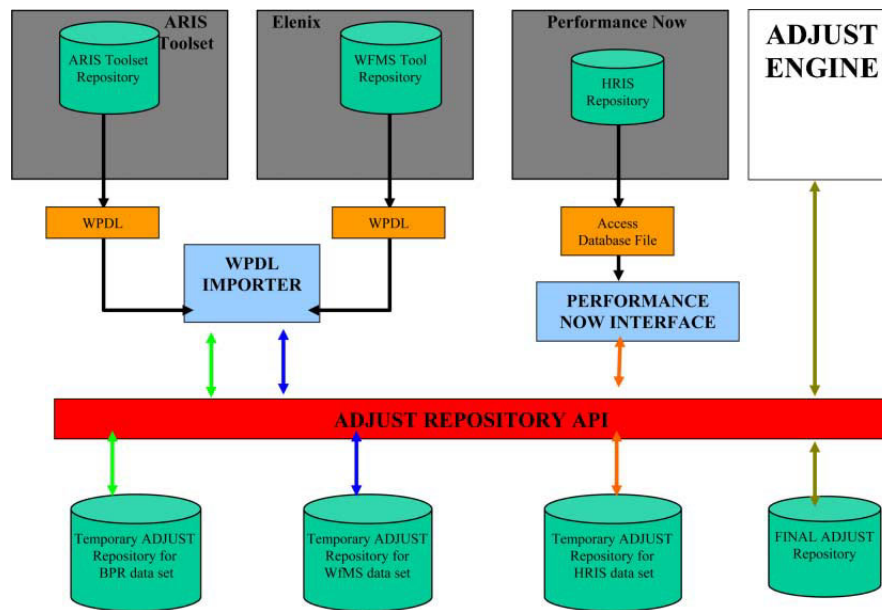


Figure 4 The Adjust Software Platform.

engine's repository in order to perform a number of analyses. The holistic meta-analysis that is performed in ADJUST engine is not performed in any of the satellite tools.

The ADJUST platform contains the following tools:

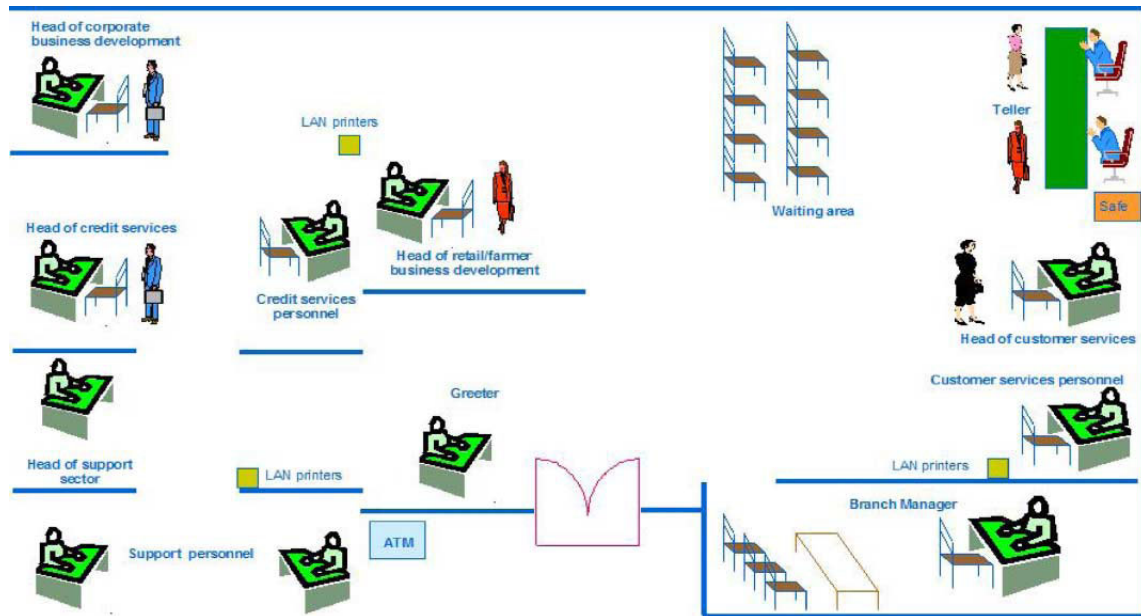
- *Satellite tools*: there are interfaces with three tools one for each domain (Process Management, Workflow Management, Human Resources). The process management tool interfacing with the ADJUST engine is the ARIS toolset (Gruhn and Laue, 2007; Bertoni *et al.*, 2009; Reijers *et al.*, 2009), that is considered one of the most dominant in its category. The HRM tool is Descriptions-Performance Now a tool that is freely available. Finally, the workflow tool we are using is called Elenix, a low cost tool available in the market. As we will explain later the ADJUST interfaces to these tools will work with any other tool in their category if it adheres to Workflow Process Definition Language (WPDL) (Dai *et al.*, 2009; Lassen and van der Aalst, 2009; Trappey *et al.*, 2009; Tsai *et al.*, 2010).
- *WPDL exporters*: tools that are specific for each satellite tool. They are to convert the information that is kept in a satellite tool repository form its internal representation to WPDL. Two WPDL exporters have been build: ARIS WPDL Exporter and ELENIX WPDL Exporter.
- *WPDL importer*: this tool is checking WPDL files for syntactical errors and then converts the information which will be imported into a temporary BPR or WorkFlow Management System repository in ADJUST. Since this tool can read and convert WPDL files then any process definition tool that export processes in WPDL can be used in ADJUST platform.
- *ADJUST repository API*: it is a set of functions stored in a Dynamic Link Library (DLL) that can be used by any other tool in order to import, export, or update information from and into ADJUST Repository. The DLL library can be used by another application in any programming environment. Any body who owns the DLL can link his/her specific tool to ADJUST Repository. The API has been used in order to build the WPDL exporter, Performance Now interface, and Unification application.
- *HRM interface*: this is an interface to an HRM tool selected (Description/Performance now). The interface in order to test API's functionality and to import information from Description/Performance Now to ADJUST Repository.
- *Adjust consistency checker*: this tool is comparing the information that exists in the temporary ADJUST Repository for inconsistencies (e.g. a record that appears in one tool it does not appear in the other, etc.). The user can receive a report with inconsistencies and he or she has to go back to the source of the information to make any corrections.
- *Adjust unification*: this tool gets the consistent information that is stored into temporary ADJUST repositories, performs unification and imports the unified data set into the final ADJUST Repository.
- *ADJUST engine*: the heart of ADJUST platform. With this tool a user can perform analysis based on the information stored in the final ADJUST Repository. This tool can be used as a stand-alone tool and for that reason it can be used for process definition tool as well. It contains a number of functionalities such as: reporting, analysis (user defined and predefined), database management, etc.

CASE STUDIES IN BANKING

The ADJUST methodology and tool was applied in a three big European Banks over a period of 10 years. All three of them wanted to restructure their activities and processes so that they are aligned with

New IT systems were installed and a new retail network with many brand new branches was constructed and made operational.

Below we present a design of the ARIS toolset for a typical would be branch in one of the case studies:

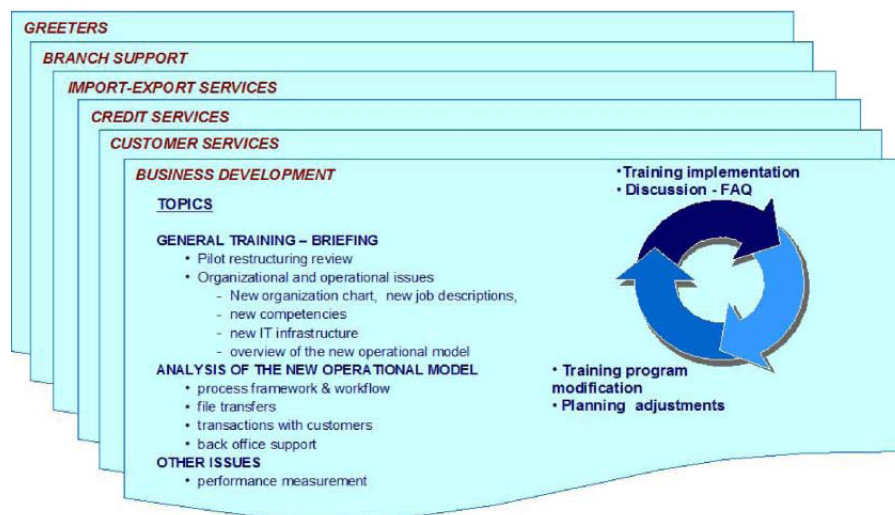


their strategic objectives and the corresponding critical success factors.

Since employee effort accounts for more than 80% of their costs it was of critical importance the redistribution of employee effort in the best possible way for the execution of the reengineered processes.

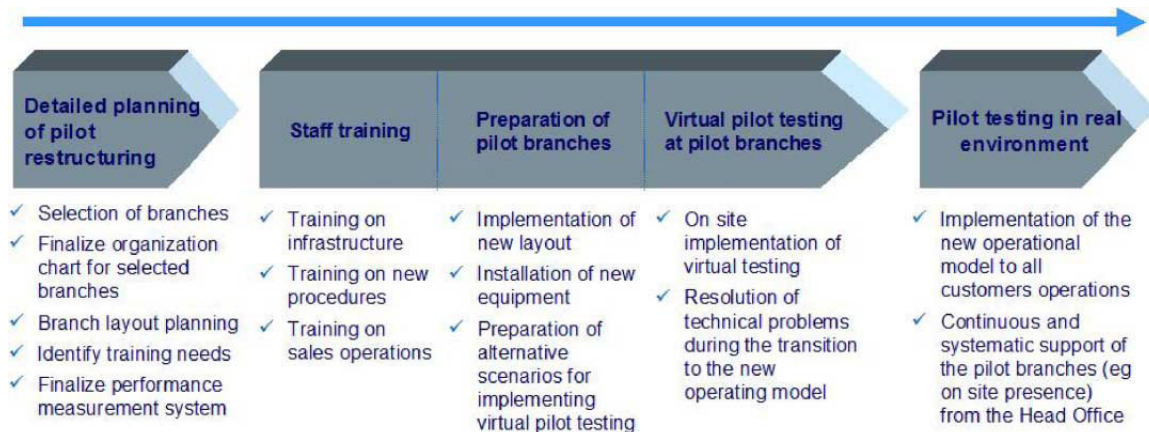
Each one of these projects lasted for more than three years and concluded in the creation and implementation of strategic and business plans.

The picture above presents job descriptions, IT system elements, spatial characteristics as well as interactions with clients. In order to restructure these organizations large education and training projects were launched. In these projects employees were trained in new processes, organizational structures, managerial systems, banking IT system, electronic banking and the new management and reporting information systems. Below we present a high level description of the training plan per job description:



Migration plans to new structures, processes information technology and human resource management systems were introduced. New systems focusing on performance measurement, training, benefits and reward systems and their links to career advancement and pay ranges per job description were also developed. In the following picture we present a migration plan who had a yearly duration.

Given the pressure to reduce costs and increase performance, there is an interest in making new business processes maximally efficient. Although much effort is expended on developing elegant and ingenious business process modelling (BPM) techniques (Rolstadas, 1998), it is often overlooked that factors concerning the process environment (e.g. culture, shared knowledge) and the agents that support the processes (i.e. people and their level of



In the next sections we are going to present the theoretical underpinnings and the metrics used in ADJUST for process, human resource management and organizational performance measurement with references made in the three case studies in the banking. Unfortunately, for confidentiality reasons we can not disclose data on the actual case studies but only indicative data and results of the application of the methodology and the tool.

HRM PERFORMANCE MEASURES AND LINKS IN ADJUST

One driver for the rise in performance measurement techniques has been the desire to cut out unnecessary organizational bureaucracy and levels of management. In the classical management model, both these roles are concerned with manipulating information for the benefit of the directorial and operational business units. With the availability of information and communication technology, the temporal and spatial foundations of business are changing and these traditional roles are also changing, or disappearing altogether (Neely, 1998; Bititci *et al.*, 2000).

For example, workers in a company can now be geographically far apart and send (electronic) documents in real-time; the associated removal of administrative layers means that core profit-making business processes and their effectiveness are being forced into a sharper focus (Hammer and Champy, 1993).

skill and knowledge) can have a significant influence on their success and acceptance (Bevilacqua and Thornhill, 1992; Carr and Johansson, 1995; Ghalayini and Noble, 1996).

Popova and Sharpanykh (2010) argue that Performance measurement and analysis is crucial for steering the organization to realize its strategic and operational goals. Relevant performance indicators and their relationships to goals and activities need to be determined and analysed. Current organization modelling approaches do not reflect this in an adequate way and that there is a need to fill the gap by presenting a framework for modelling performance indicators within a general organization modelling framework.

Gregoriades and Sutcliffe (2008) argue that Human performance constitutes an important parameter to business process performance and should be modelled based on Human Performance Shaping Factors (PSF) and assessed using Bayesian Belief Networks (BBN). Jazayeri and Scapens (Jazayeri and Scapens, 2008) have created the Business Values Scorecard in their performance measurement system on a culture change project, which focused on five key values: performance, people, customers, partnerships, innovation and technology.

These HRM related data items in ADJUST are included in job descriptions links job descriptions with a competency framework that specifies career planning and individual performance measurement.

The typical measures are related to:

- Soft factors.
- Knowledge.

Soft factors

Organizations should not be viewed and assessed within a quantitative, structural framework, with poor integration of the 'soft' people and cultural issues. The soft factors framework described here provides a set of measures of the factors influencing employee and team performance. The performance model is based on the interactions between people and organizational culture (Longenecker and Fink, 2001; Flapper *et al.*, 1996).

The objectives and, by consequence, benefits of the soft factors in a job description are:

- ✓ Clarification of the links between hard factors such as costs, and soft 'people' issues such as culture.
- ✓ Facility for tracking soft factors, performance and variations over time.

The aim here is to assess measures related to:

- People factors such as staff turnover and motivators.
- Culture of the organization in terms of the informal micro-culture and the formal macro-culture.
- Investment in the employee-team and its environment for improved performance.
- Performance of the employee-team, such as the quality of decisions or actions.
- Benefits of the investment and improved performance.

- Time over which the investment takes place, and performance is tracked.

Time is treated an implicit factor, while the environment of the employee-team, or the team context, is accommodated within the culture factors. Micro-cultural factors reflect the way things are done in the immediate team, the local environment of the team and the overall organizational environment. Macro-culture reflects wider organizational issues such as monetary reward.

The soft factors in ADJUST included in the job descriptions are the following:

- (1) Customer orientation.
- (2) Results orientation.
- (3) Planning and organization of work.
- (4) Relationships with other employees
- (5) Decision making.
- (6) Problem solving.
- (7) Team work.
- (8) Innovation-flexibility.
- (9) Communication skills.
- (10) Motivational skills.
- (11) Leadership.
- (12) Commitment.

The values for the subjective measures presented above take values in the scale from 1–10. The data for all these measures are provided from an interface or could be inserted manually by the user of the ADJUST system. No database data are required by the system for the evaluation of these measures.

The results for every measure will be displayed in a graph with two axes. In the one axis will be the scale and in the other the job title, department, measure, etc. Graphs provide comparisons of Planned (job description level) vs. Actual employee level values as shown in the Figure 5.

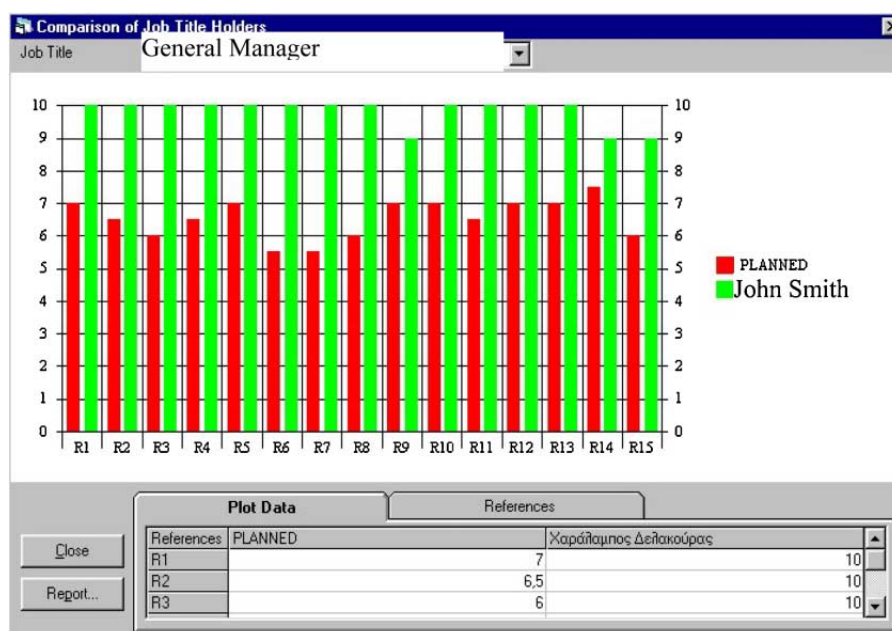


Figure 5 Planned vs Actual Employee Performance Data.

Graphs can also be produced at the company Level, per Department, or for Comparison between different job titles, different employees and different business processes.

Knowledge management and business process management

State of the art

Knowledge is defined as the daily sharing of operational knowledge. Knowledge, like other production resources, adds value to the organization. Many researchers have linked organizational knowledge creation to business strategy and organizational performance (Davenport and Prusak, 1998; Glazer, 1998; Davenport, 1999; Connelly *et al.*, 2000; Kwan and Balasubramanian, 2003; Lee *et al.*, 2005; Tsai and Li, 2007; Andrade *et al.*, 2008; Tseng, 2008; Han and Park, 2009; Li *et al.*, 2009; Zheng *et al.*, 2009; Yang *et al.*, 2010b; King *et al.*, 2008).

Most of these researchers concentrate on the link between business processes and knowledge creation, capture, retrieval and the distribution. Others like Chang *et al.* (2009), Lee *et al.* (2005) and Tseng (2008) have created frameworks that assess and measure knowledge readiness and capture in organizations. Kwan and Balasubramanian (2003) have created a framework called KnowledgeScope an integrated workflow support capability that captures and retrieves knowledge as an organizational process proceeds. Kock *et al.* (2009) conducted a study of a large number of completed BPR projects and have concluded that communication flow in the business process amongst the process participants and external stakeholders is the most critical factor for success. They also concluded that communication flow is very closely coupled with process model quality and its link with the reality and real time execution.

Zheng *et al.* (2009) argue that knowledge management plays a critical mediating role in linking concepts like organizational culture, structure and effectiveness. Niemi *et al.* (2009) have linked the knowledge management with skills, roles and responsibilities, performance measurement the rewards and incentives systems.

In ADJUST we take into consideration the outcomes of previous research, we recognise the importance of knowledge management and propose a new approach regarding the link amongst knowledge management, process orientation, workflow, communication flow, organizational structure etc. Knowledge is handled both at the holistic (strategic and process level) as well as at the individualistic level of job descriptions with workflows and communication flows being described not only at process level but also amongst job descriptions (structural concepts). Finally all these concepts are used in performance measurement and rewards systems.

Explicit knowledge and tacit knowledge

In order to capture knowledge in BPR, knowledge management concepts have to be integrated in business process modelling (Nunes *et al.*, 2009). There are two knowledge types that are fundamental to an understanding of operational knowledge, i.e. explicit (or articulated) and implicit (or tacit) knowledge (Lubit, 2001). Tacit knowledge (Boiral, 2002) is difficult, though not impossible, to articulate, and is mostly heuristic, subjective and internalised. It is learnt through concrete examples, experience and practice. In contrast, explicit knowledge can be formally represented and communicated through languages. However, knowledge in general is dynamic and represents a capacity to act and adapt within a changing environment; so, although explicit knowledge can be represented, it cannot be exhaustively detailed in statements of belief or prescriptions for behaving. Hence, tacit and explicit knowledge are not mutually exclusive. All knowledge relies on being tacitly understood and applied. But tacit knowledge can be made explicit by externalization, just as explicit knowledge can become tacit by internalization. Tacit knowledge can also be communicated from one holder to another without becoming explicit (Liebowitz and Wright, 1999).

Agents and objects as knowledge carriers

Recently agent based systems have appeared both in knowledge management and workflow research literature (Leonardi *et al.*, 2007; Yang *et al.*, 2010a). There are two distinct types of knowledge carriers, namely agents (individuals and groups of individuals) and objects (inanimate things such as manuals, drawings, logbooks and computer records). Codified knowledge is the knowledge that is recorded in an explicit format, in text, pictures or speech. It is largely contained in objects. Although it is called knowledge, it is really encoded data and information, which requires human reasoning to be made sense of and effectively utilised. Uncodified knowledge, on the other hand, represents tacit knowledge that resides in human agents.

But when knowledge is only held in individuals, it is localised: it must be shared by the individuals working together on an operation to be effectively utilised and to become a basis for collective or organizational learning.

Knowledge that resides in agents is uncodified and primarily tacit. Part of it can be conceptualised and formalised. The knowledge of agents includes the rules for doing certain tasks and the formulae acquired through experience and learning. This knowledge then becomes increasingly explicit and can be transmitted through formal languages or speeches, and so be externalised and codified. Although such rules and formulae are explicit, they still require tacit understanding, as the context and the cognitive reasoning underlying the knowledge cannot be made explicit.

This tacit understanding is transferred through informal processes such as social networks, mentoring and coaching. In a process operation, both formal and informal communications are important for the transfer of such tacit knowledge.

Knowledge that resides in objects is codified and explicit. Such objects are critical in process operations because there must be explicit knowledge or common information that the production team can refer to and adhere to. The knowledge content of such carriers must be clear and unambiguous, so that it can serve as a platform for common tacit understanding and cognitive interpretation. Knowledge objects such as a night order books, area log books, written safety procedures, must be made available for knowledge sharing in the operational environment. The quality and accessibility of such knowledge objects will affect not only knowledge sharing amongst the teams but also collective learning. To ensure effective sharing of knowledge, it is necessary to establish prioritization, categorization and consolidation of information.

Declarative knowledge and procedural knowledge

ACT theory distinguishes two types of knowledge involved in performing a specific task: declarative knowledge and procedural knowledge. Declarative knowledge is essentially explicit and covers the facts and rules relevant to the task (Ndlela and du Toit, 2001).

On the other hand, procedural knowledge is mostly tacit and is concerned with knowing how to perform the task effectively. According to ACT theory, all knowledge is initially presented in the declarative form. The process of acquiring expertise or specialised knowledge occurs through knowledge conversion from declarative form into procedural form. This conversion takes time and occurs through a variety of learning mechanisms such as composition and proceduralization.

Running a process operation involves human operators and other functional agents. The presence of the human dimension means that the traditional mechanistic or reductionistic approach is inadequate for describing the system.

Organizational learning

Many researchers have focused on the link between knowledge management and organizational learning [see for example collection of papers in Liao and Wu, 2010]. According to Liao *et al.*, (2010) knowledge management is closely related to organizational innovation and learning. Human activity systems embody inherent incoherence such as ambiguities, uncertainties and contradictions. These have to be harmonised for knowledge to be shared and organizational learning facilitated. Therefore it is necessary to pay attention to contextual influences such as social, cultural and organizational factors, which affect knowledge sharing and organizational learning.

Hannah and Lester (2009) have developed a multilevel approach for building and leading learning organizations which is based on the individualistic (micro), network (meso) and systems (macro).

Liao and Wu (2010) have very added a philosophical foundation to knowledge management and introduced the term 'inquiring organization' in organizational learning via knowledge management. They argue that if a solid knowledge management system is in place then employees are eager to learn and explore in order to improve their decision making abilities.

Value creation and intellectual capital.

Many researchers argue that capturing knowledge via organizational learning leads to value creation and intellectual capital (Wu and Cavusgil, 2006; Lai *et al.*, 2009; Hsu and Fang, 2009; Yang *et al.*, 2010b). Lai *et al.* (2009) have concentrated on the link of information capital and organizational learning and how this may affect information behaviour and create information value. Wu and Cavusgil (2006) have highlighted the importance and the joint value created when organizational learning is used in interfirm relationships. Yang *et al.* (2010b) have introduced knowledge creation strategies for the creation of 'organisational knowledge assets'.

The value created in information and in interfirm relationships has inspired researchers to investigate the link between organizational learning and intellectual capital (Hsu and Fang, 2009). An issue that is very important for knowledge based firms for years (Liao *et al.*, 2010).

Value creation and intellectual capital has been a significant issue in Management Accounting research in recent years. Mouritsen and Larsen (2005) argue that we are now experiencing the second knowledge management wave. The intellectual capital wave, with the first being focused on the individual, in which knowledge resources are represented in intellectual capital information in charts of accounts as 'intellectual capital statements'.

In ADJUST the several types of knowledge are measured. Some indicative include cultural, organizational, process, team based, process, product, technology and innovation, etc. These knowledge measures are included in job descriptions:

- (1) Professional knowledge (degrees)
- (2) Market knowledge
- (3) Knowledge of culture and customs
- (4) Product knowledge
- (5) Process knowledge
- (6) Technology expert knowledge
- (7) Technology user knowledge
- (8) Specialist equipment knowledge
- (9) Foreign language knowledge
- (10) Team work knowledge
- (11) Management discipline knowledge

The ADJUST system provides similar graphs as in the case of soft factors.

CAPTURING DECLARATIVE KNOWLEDGE THROUGH WORKFLOW TRANSACTIONS

In many cases process reengineering is not linked with other HRM exercises in the organization resulting in total confusion at the level of the employee who participates in business processes with specific process objectives defined by strategy. In many cases these process objectives are conflicting with the objectives of his/her job descriptions, performance evaluation and career path that is defined by HRM. In other cases employees utilise workflow management technology through workflow process models, transactions etc. that are not necessarily in accordance with the business models described in BPR and responsibilities, performance measures etc. described in HRM (Bititci *et al.*, 1997; Davidson, 1993). Reijers and van der Aalst (2005) describe the need for integration between workflow and BPR concepts methodologies and tools.

In Adjust we integrate job descriptions, process and workflow elements with the aim of capturing the declarative-explicit knowledge of Agents in the organization we have included the following fields in job descriptions.

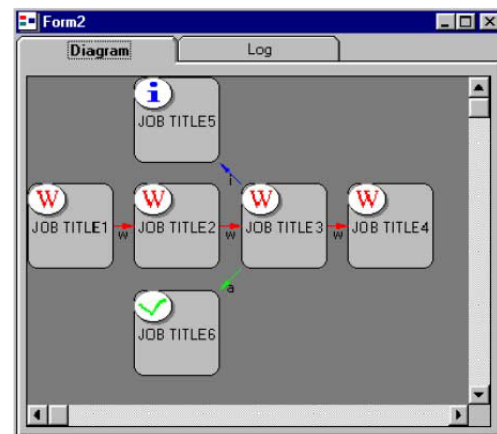
Main Job Title	This field can either be the sender or the recipient, depending on the value of the <i>Sender</i> field
Activity	This field holds the task that the sender carries out
Transaction	This field holds the transaction name that specifies an object that flows between agents (humans or software agents) in the organization.
Other Job Title	This field can either be the sender or the recipient, depending on the value of the <i>Sender</i> field
Sender	If this field is checked, then the <i>Main Job Title</i> is the sender in this job description. If not, then the sender is the <i>Other Job Title</i>
Work/ Information/ Approval	This field indicates the purpose of the job description

A transaction is an object that is exchanged between agents in the organization. Agents are mainly human but can also be software agents. An example of a transaction could be an architectural design that is exchanged between Architects and civil engineers before it takes its final form.

At the system level details about transactions are inserted as shown in the following figure. The declaration in the record explains that JOB TITLE 2 receives TRANSACTION1 in the context of ACTIVITY1 from JOB TITLE 1). The 'Sender' check

button remains unchecked in order to declare that JOB TILE 2 is a receiver of this transaction.

By repeating the same steps in various job descriptions for the same transaction we can produce the 'transaction trace' as shown in the following figure:



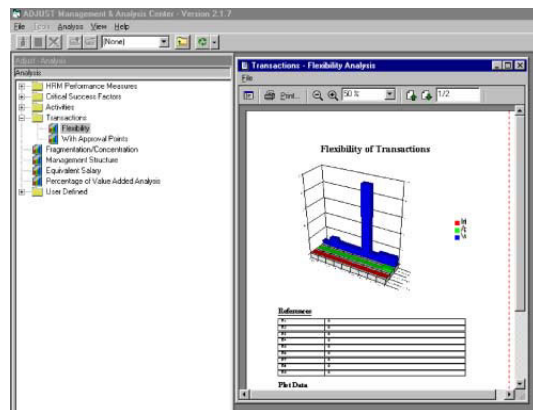
This transaction trace declares that there is a work-flow (rectangles marked with W) from job title (agent) 1 to 4. Additionally, job title 3 sends his result (object) for approval to job title 6 (rectangle marked with ✓) and when he receives the final approval he sends the final approved result to job title 5 for informational (rectangle marked with I) purposes.

A transaction trace is constructed for each transaction. The more approvals associated with the transaction the more control we impose on it. The less approval points the more flexible the transaction. Typical analysis results that will be produced by the system are the following: A similar kind of analysis will be performed at the level of a process, department and job title with more focus on the approval points. The calculation of the approval points per process or department will be performed on the basis of the job description where there is a declaration of approvals per transaction per activity

The system produces analysis graphs which contain:

- The number of information exchanges for each transaction (red line)
- The number of approval points for each transaction (green line) and
- The number of works steps of each transaction (blue line)

as shown in the following figure:



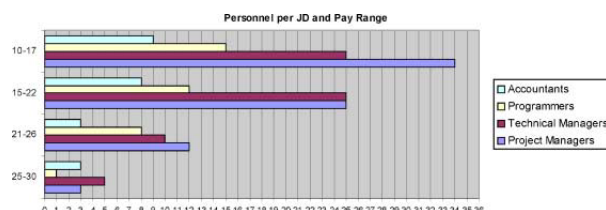
FROM SOFT FACTORS AND KNOWLEDGE TO PAY RANGES

A pay range is provided per job title (manually computed by the user) depending on the overall Soft Factors and Knowledge Level as well as other targets related to critical success factors, process achievements and *experience*. The user can provide pay ranges and corresponding salary values. For example:

Pay Range	Min	Max	Factors
1	10 000	15 000	List of measures for the employee to belong to this pay range (in Knowledge, Soft factors and other areas)
2	13 000	17 000	List of measures for the employee to belong to this pay range (in Knowledge, Soft factors and other areas)

The system provides analysis graphs like job-titles and pay ranges as well as job titles and average salaries (planned and actual) as well as actual salary and number of people that correspond to this salary.

Typical examples of business analysis results include:



ORGANIZATIONAL STRUCTURE RELATED PERFORMANCE MEASURES

Management structure analysis focuses on evaluating structural efficiency and leveraging of management resources.

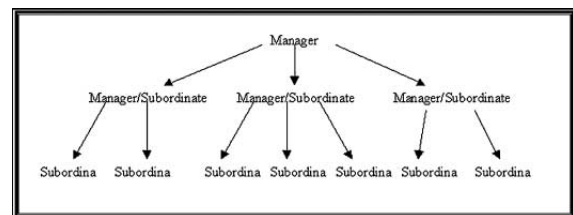
Management structure analysis focuses on evaluating structural efficiency and leveraging of management resources. There are several key calculations that play a significant part in this kind of analysis:

Span of control

- This is how many employees a manager supervises excluding secretaries (or administrative support personnel).

This analysis calculates the span of control in an organizational structure.

- (1) For each manager find the manager/subordinate ratio. Each subordinate might be a manager.



- (2) Calculate the Average Ratio of manager/subordinate throughout the whole organizational structure

$$\text{Average Ratio} = \frac{\text{Sum}(\text{manager/subordinate ratios})}{\text{Number_of_manager/subordinate_occurrences}}$$

This analysis calculates the span of control for a specific organizational structure (i.e. company, division, department, employee).

- *Company span of control*: the analysis takes into consideration all the employees in the organization starting from the employee that has no supervisors and working its way down to lowest level of the organizational structure (i.e. employees with no subordinates).
- *Division span of control*: the analysis calculates span of control for employees that belong to a specific division starting from the employee in that division that does not have any supervisors and working its way down to the lowest level of employees, which do not have subordinates.
- *Department span of control*: the analysis calculates span of control for employees that belong to a specific department starting from the employee in that department that does not have any supervisors and working its way down to the lowest level of employees, which do not have subordinates.
- *Employee span of control*: the analysis calculates span of control for a specific employee starting from that employee and working its way down to the lowest level of employees which do not have subordinates.

Cost to manage

This represents the manager's salary cost spread across all subordinate salary costs.

For each organizational entity or organization as whole
Find his/her subordinates
Loop

$$\text{Cost_to_Manage} = \frac{\text{Salary_of_managers}}{\text{Sum}(\text{Salary_of_subordinates})}$$

Calculate the Average Ratio of manager/subordinate throughout the whole organizational structure
Average Ratio

$$= \frac{\text{Sum}(\text{Cost_to_Manage})}{\text{Numbe_of_manager/subordinate_occurrences}}$$

This analysis calculates the cost to manage ratio for a specific organizational structure (i.e. company, division, department, employee).

Layers of management

This represents the number of management layers in the organization. This analysis counts the depth of managers who report to other managers.

For each organizational entity or organization as a whole
Repeat until there are no subordinates
Find those who do not report to anybody
Put them in Level 0
For those who have been put in Level 0
Find the subordinates and put them in Level 0 + 1
Level 0 = Level 0 + 1

Loop

Layers of management = number of Levels

This analysis calculates the layers of management ratio for a specific organizational structure (i.e. company, division, department, employee).

PROCESS RELATED PERFORMANCE MEASURES AND LINKS

Activity based costing for calculating process costs have been used extensively (Curtis *et al.*, 1992; Tatsiopoulou and Panayiotou, 2000; Tornberg *et al.*, 2002; Askarany *et al.*, 2009; Johnson and Kirchain, 2009a; Johnson and Kirchain, 2009b). All of these approaches attribute costs to all cost drivers related to a process resource consumption. One of these resources is the human resource.

In ADJUST we concentrate only on the human resource related costs. For this reason we focus on employee effort devoted to activities. Based on the effort per employee per activity we calculate the full time equivalents per activity. Which is the sum of employee's effort divided by 100.

$$\text{FTEs per Activity} = \frac{\sum (\text{EffperAct})}{100}$$

The calculation of FTEs per activity is a very important activity characteristic as it shows the number of employees that could be dedicated to this activity on a full time basis.

Activity cost calculation and analysis

This analysis calculates the total activities cost for a specific level of activities. The equation for calculating the cost of an activity is as follows:

$$\text{ActCost} = \sum (\text{EmpSal} \times \text{EffperAct})$$

where

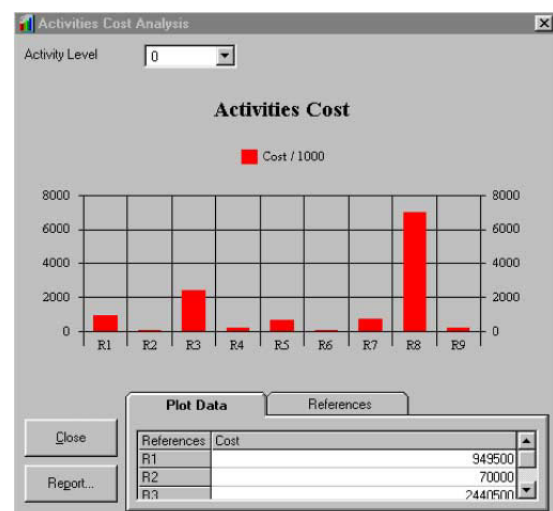
ActCost = the total activity cost

EmpSal = employee salary

EffperAct = employee effort for a specific activity

The analysis returns the cost for each activity that belongs to a selected activity decomposition level. The decomposition level is automatically calculated during the creation of the analysis. The user can select any decomposition level.

Example results are presented in the following diagram:



Link between strategy, processes and employees' effort

The link between strategy, business processes and other associated concepts like technology fit, flexibility etc. has been the focus for many researchers in recent years. Arias-Aranda *et al.* (2010) have created a fuzzy expert system based on a fuzzy rule set that elaborates on the link between strategy, flexibility and performance. Trkman (2010) argues that although business process management is a popular concept, it has not yet been properly theoretically grounded. This leads to problems in identifying both generic and case-specific critical success factors of BPM programmes. The paper proposes an underlying theoretical framework with the utilization of three theories: contingency,

dynamic capabilities and task–technology fit. The main premise is that primarily the fit between the business environment and business processes is needed. Then both continuous improvement and the proper fit between business process tasks and information systems must exist.

In adjust we follow this identification and we attempt to create a strong and sustained linkage between strategy and the way work is done is an enduring challenge in complex organizations similar to (Werbel and DeMarie, 2005). Business processes in BPR define how work is done and must be redesigned in a way that is consistent with organizational strategy.

In ADJUST a technique called Mission/Non-Mission analysis is used, in order to analyse the extend to which agents' activities and organizational processes are secondary to the attainment of the organizations objectives or mission. The major benefit of this analysis is the ability to isolate and quantify non-missionary effort in selected work groups as well as the probable causes of it. An activity can be classified as non-mission according to the following criteria:

- The activity is organizationally misplaced.
- The activity is being duplicated because of a 'do it by ourselves is better' attitude.
- The wrong skill level is being applied to the activity.

Performance measurements in the strategy perspective

Many BPR Methodologies and Tools have highlighted the importance of linking business objectives to activities so that discrepancies between corporate strategy and employee effort and activity costs are identified and corrective actions are undertaken.

Most methodologies introduce the Balanced Scorecard as the most appropriate for analysing business performance (Thompson and Mathys, 2008; Vergidis *et al.*, 2008; Tseng, 2010; Yüksel and Dağdeviren, 2010). Performance measures in the four dimensions of the Balanced Scorecard (BSC) are associated with business processes the performance of which is critical for the achievement of these goals.

Many researchers are recently using the fuzzy Hierarchical or Analytic Network Process techniques as an analysis method in BSC (Thompson and Mathys, 2008; Dağdeviren and Yüksel, 2010; Tseng, 2010; Yüksel and Dağdeviren, 2010).

Tseng (2010) argues that although there is an abundance of literature on the BSC framework, there is a scarcity of literature regarding how the framework with dependence and interactive relationships should be properly implemented in uncertainty. He proposes a hybrid approach which combines the Analytic Network Process to analyse

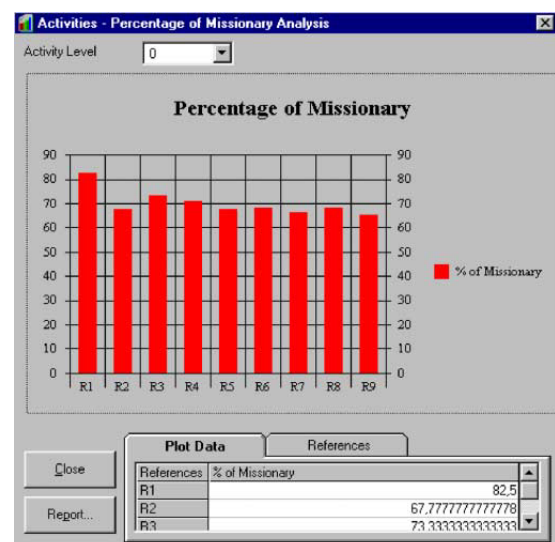
the dependence aspects and the fuzzy set theory to evaluate uncertainty.

Other researchers use Strategy Maps as a means of analysing BSC performance measures (Quezada *et al.*, 2009; Buytendijk *et al.*, 2010). In our view the best framework for representing and analysing BSCs is the fuzzy cognitive maps which combine fuzziness to analyse uncertainty, mapping capabilities, analysis engine and simulation and map interconnections (Glykas and Xirogiannis, 2004; Xirogiannis and Glykas, 2004a; Xirogiannis and Glykas, 2004b; Xirogiannis and Glykas, 2006; Xirogiannis *et al.*, 2008; Wu *et al.*, 2009).

Mission–Non-Mission analysis is an extension to the critical success factors importance analysis. The following type of table is used for relating critical success factors to activities:

	ACT1	ACT2	ACT3	ACT4...	Total
CSF 1	Scale from 1–10				
CSF 2					
CSF 3					
CSF 4					
Total					

The analysis on the above table is based on the degree of importance of each Activity and the level of achievement of the critical success factors via the activities. An Activity is considered as being very important (missionary) if it contributes to more than 50% of its total contribution potential. The total contribution potential is calculated by multiplying 10 (the highest grade in the scale) by the number of critical success factors. In the case we have 4 CSFs the contribution potential is $10 \times 4 = 40$. In this case, if the sum of the contribution in each CSF is more than 50 then the Activity is considered as very important. Example reports are presented in the following diagram.



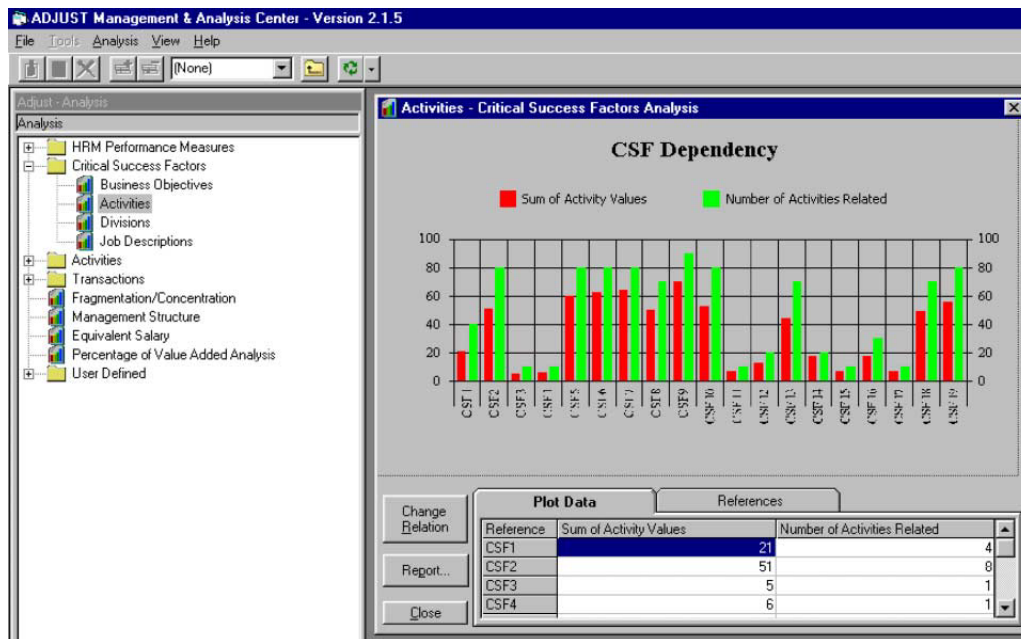
The algorithm behind the analysis is as follows:

```

For each Activity
  Sum all CSFs values
If Sum < Threshold then Activity is non-Mission
If Sum > Threshold then Activity is Mission
Loop
Threshold = number of CSFs × highest CFS value × 0.5

```

The analysis will return results only if the user has associated activities with critical success factor values. Example reports produced by adjust are presented in the following diagram:



In a similar manner a CSF has a highest level of achievement if all Activities are associated with it and take a number of 10 in the scale. If the sum of the Activities is more than 50% of its achievement potential then the CSF is considered as multiply achieved.

Activity value added analysis

In this analysis the focus is on the determination of each activity's (or set of activities) value to the 'customers' and to the organization as a whole.

Activities can be classified into three primary types:

- **High value added activities:** These activities are critical to the achievement of existing organizational strategies and have high impact on product quality and customer satisfaction.
- **Business value added activities:** These activities are critical to the attainment of general business operations and usually support high value added activities.
- **Low value added activities:** Activities, which provide low or negligible overall value from product, customer and organizational strategy perspective.

The classification of each activity to each one of the above three categories is subjective and presented by the user in a scale from 1–10.

Fragmentation/concentration analysis

Fragmentation is the degree to which effort applied towards an activity is dispersed within a unit, a department, or the entire organization. It is expressed in terms of the number of

employees it takes to generate one FTE (full time equivalent) and is calculated in the following ratio:

$$\text{Fragmentation} = \frac{\text{Number of Employees that have declared a \% of effort to the Activity}}{\text{FTE}}$$

the higher this ratio, the higher the degree of fragmentation.
where

$$\text{Full Time Equivalent (FTE)} = \frac{\sum \text{effort_of_employees_in_activity}}{100}$$

Concentration is the inverse of fragmentation. It is the average amount of time employees spend on a particular activity and is expressed in the following ratio:

$$\text{Concentration} = \frac{\text{FTE}}{\text{Number of Employees that have declared a \% of effort to the Activity}}$$

The lower this index, the lower the concentration or the higher the degree of fragmentation.

This analysis calculates the fragmentation and concentration values of activity in a specific activity decomposition level. The calculation algorithm is shown below:

For each Activity

Full Time Equivalent (FTE) = Sum (effort of Employees in Activity)/100 = Full Time Equivalent

Concentration = FTE/Number of Employees that have declared a % of effort to the Activity

Fragmentation = Number of Employees that have declared a % of effort to the Activity/ FTE

Loop

The analysis will return results only if the user has associated activities with employees and efforts.

Equivalent salary analysis

The purpose of equivalent salary analysis is to determine if work can be performed at a lower cost to the organization, either through internal re-allocation to less expensive employees or

This kind of analysis (total activity cost divided by the total FTEs allocated to the activity) is appropriate for:

- Identifying trivial activities that are performed by highly qualified employees and vice versa.
- Evaluating whether compensation for activities, products or services is competitive with the marketplace, as in weighing a decision to out-source a particular product or service.

The analysis calculates the equivalent salary of a specific department for an activity decomposition level. The calculation algorithm is shown below:

For each Activity

For each employee that has declared % of effort to the Activity

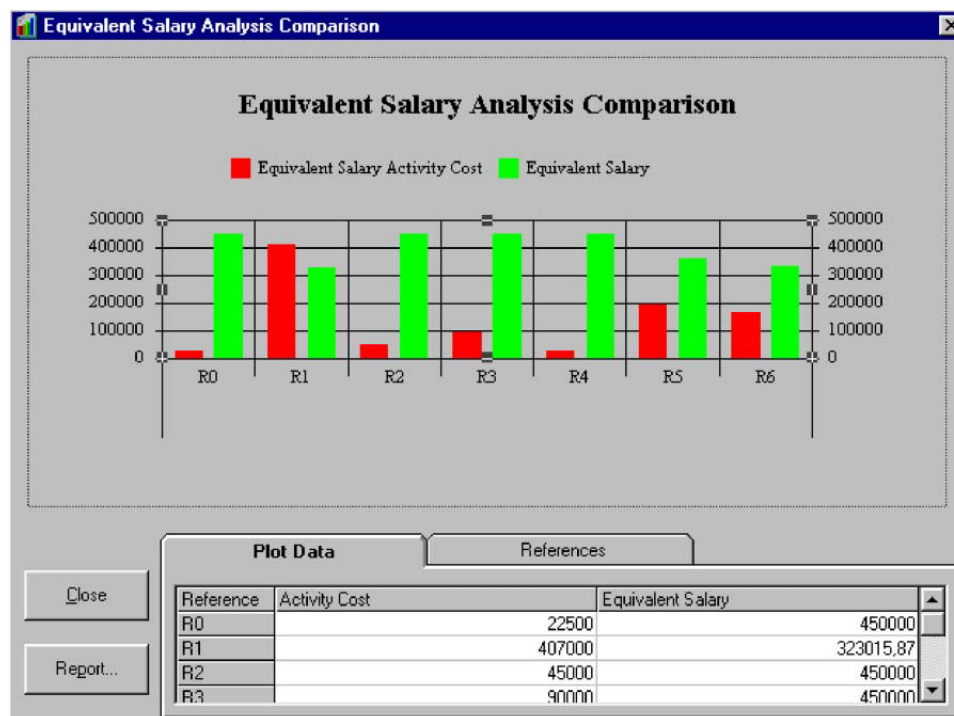
Activity Total cost = (Salary × % of effort)

Equivalent salary = Activity Total cost/FTE

Loop

Loop

The analysis takes into consideration all the employee salaries that belong to a specific department as well as their declared effort for a specific activity decomposition level. Example reports are the following:



through outsourcing. It is expressed in the following ration:

$$\text{Equivalent Salary} = \frac{\text{Total Activity Cost}}{\text{Activity FTEs}}$$

Process cycle time analysis

This analysis identifies which steps in a business process are critical to the timely completion of the process. By mapping work step interdependencies

the analysis identifies a (critical path), a set of work steps which, when shortened, will compress the process duration. This kind of analysis is often iterative.

When a work step is shortened, a resultant critical path may include a different set of work steps. This path is repeatedly refined as part of the overall redesign process. When the critical path in a process is identified the improvement of the efficiency of any individual work step in the critical path might result in big improvements on the efficiency of the process.

In ADJUST this kind of analysis is particularly useful when analysing a complex process that has multiple interrelated steps of sub-processes being executed concurrently.

CONCLUSIONS AND FURTHER RESEARCH

A holistic performance measurement methodology and toolset for BPR was presented. The aim is to integrate existing business modelling, human resource management and workflow systems via interfaces both during organizational design and in real time. Business analysis is performed with a series of metrics and techniques both quantitative and qualitative mainly based on effort based costing.

Different analysis techniques from various disciplines are applied in different perspectives. Most of these techniques are influenced from organizational theories, strategic planning, accounting, information system development, transaction cost economics and existing work in process management.

The application a smaller subset of techniques like flexibility and dimensions of transactions are novel to performance measurement. Other techniques like management structure analysis, concentration analysis, equivalent salary analysis and fractionalization analysis are used from a different viewpoint.

The major contribution of ADJUST in business analysis is the ability to analyse the resulting business model much more holistically compared to other methodologies and tools.

Business analysis in ADJUST allows users to:

- Analyse the organization from both the holistic and individualistic view.
- Explore the relationship between organizational structure and processes.
- Examine the alignment between business strategy, business processes and employee objectives.
- Examine the level of communication, coordination and control in existing structure and processes.
- Analyse the impact of creating a market metaphor for organizational behaviour.

- Analyse the level of effectiveness of activities and processes.
- Analyse the level of efficiency of activities and processes.

Applications of the ADJUST methodology and toolset in a variety of case studies help companies measure and evaluate the efficiency of their standard software-supported processes.

ADJUST has a broad range of applications for implementing sophisticated process controls.

The information and evaluations provided support at:

- Continuous improvement of processes, both within the company and across companies.
- Weak point analysis during process engineering.
- A basis for benchmarking projects, both within the company and across companies
- Outsourcing decisions.
- Self-assessment by process participants.
- A basis for automated early-warning systems.
- Collecting data for activity-based costing and simulation.
- Calculating the efficiency and profitability of standard software implementations.

Further research

ADJUST has been applied in various business sectors like banking, production and manufacturing, telecoms, information technology, etc. The results of case studies as well as user's views on the techniques and ratios used will be integrated into the methodology and will help in adding new techniques and ratios and even eliminate others that are not considered as useful.

There is an immediate need for further research in the area of ratio and business concept interlinking which will help users of ADJUST benefit from predefined links amongst the ratios used. For example in organizational analysis the lower the span of control ratio or percentage the lower the cost to manage and vice versa. The lower the span of control the more standardized the work and processes etc. In order to resolve this problem and provide a sound solution for ratio and business concept interlinking we have explored the possibility of using Fuzzy Cognitive Maps (FCMs) (Glykas and Xirogiannis, 2004; Xirogiannis *et al.*, 2004; Xirogiannis and Glykas, 2004a; Xirogiannis and Glykas, 2004b; Xirogiannis and Glykas, 2006) in business modelling.

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